

Zeke Mohammed

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Electrical engineer with hands-on experience in embedded systems, PCB design, FPGA development, and 400–800 V DC high-voltage validation across automotive, energy, and research applications.

Education

Fordham University

B.S. Engineering Physics, Electrical Concentration; Minor in Mathematics

Bronx, NY

May 2026

- Completed degree in 3 years while working 15–20 hours/week in research and teaching roles; Sarah Colman Klemmer Research Fellow, Sigma Xi Honor Society, CSTEP Scholar

Technical Skills

PCB & Embedded: KiCad (schematic, layout, Gerber, BOM), STM32, ESP32, Raspberry Pi, Arduino, C/C++, bare-metal programming

FPGA: Xilinx Artix-7, Verilog, Vivado, pipelined architectures, fixed-point arithmetic

Software: Python, MATLAB, Java, Git/GitHub, Linux, Flask, NumPy, pandas, matplotlib, L^AT_EX, SQL

Protocols: CAN bus (ISO 11898), I2C, SPI, UART, PWM, OBD-II (ISO 15765-4)

Hardware & Test: Oscilloscope, multimeter, signal analysis, SMD/through-hole soldering, PCB assembly & rework

CAD: Fusion 360, SolidWorks, KiCad 3D viewer, Blender, 3D printing (FDM/ABS)

Technical Projects

Yamaha ATV CAN Bus Sensor Node | *KiCad, STM32, CAN Bus, Fusion 360*

March 2026

- Designed 2-layer PCB (60×40 mm, ENIG) to production intent: STM32F103 + SN65HVD230 CAN transceiver; aggregates throttle position, coolant temp, wheel speed, and battery voltage into a CAN 2.0B frame at 500 kbaud / 10 Hz; deliverables include schematic, Gerbers, and interactive BOM (21 items, \$15.10 prototype cost)
- Injection-moldable ABS enclosure in Fusion 360: IP67-ready O-ring groove, M6 ATV frame flanges, DB9 strain relief; <\$1/unit at production volume

FPGA Real-Time Image Processing Accelerator | *Verilog, Vivado, Python*

June 2025 – May 2026

- 7-stage pipelined convolution engine on Xilinx Artix-7 (Basys 3, 100 MHz) with systolic array and Q8.8 fixed-point arithmetic; 48–53× throughput improvement over single-threaded NumPy baseline (240×240 Sobel+Gaussian) at 0.85 W; three clock domains with CDC synchronizers, BRAM line buffer, Python-UART verification and VGA output

Flight Controller + Power Distribution | *STM32, C, PID Control*

November 2024 – May 2025

- Bare-metal STM32F405 firmware: 1 kHz cascaded PID loop with complementary filter sensor fusion (MPU6050 I2C/SPI); <2° RMS error, 655 μs loop time; 11.1 V 3S LiPo, four 20A BLHeli-S ESCs, overcurrent protection

Multi-Sensor DAQ System (M3) | *Python, Raspberry Pi, Flask*

October 2023 – April 2024

- Integrated 12+ sensors across CAN/I2C/UART at 50 Hz with real-time Flask telemetry dashboard; validated <2% error over 50+ hours of field testing

Professional Experience

Electrical Engineering Intern

June 2024 – August 2024

Hyzon Motors

Rochester, NY

- Characterized 200 kW hydrogen fuel cell stack via V-I polarization curve tests; Python post-processing to isolate ohmic, activation, and mass transport losses; HV validation (400–800 V DC): hipot, insulation resistance, and continuity testing on HV harnesses
- Monitored BMS telemetry over CAN bus during fuel cell–battery hybrid integration testing; logged and analyzed state-of-charge and thermal data to support system validation

Undergraduate Researcher, Lattice QCD

August 2024 – May 2026

Fordham University Physics Department

Bronx, NY

- Built Python framework for SU(2) lattice gauge theory (8³×20 lattice): 81,920×81,920 sparse Wilson-Dirac operator, BiCGSTAB/GMRES solvers, 200+ Monte Carlo configs; generalized SU(2)→SU(N) with 3× runtime improvement
- Contributing editor, *High Energy Physics Essentials* (Wiley, 2025) — Dr. Christopher Aubin; presented at CSTEP Symposium

Teaching Assistant, Lab Instructor & Lab Manager

August 2023 – May 2026

Fordham University

Bronx, NY

- TA: instruct 30+ students in experimental physics; evaluate 120+ lab reports/month with technical feedback; Lab Manager: 50-workstation teaching lab, 99%+ uptime, 40% reduction in deployment time by scripting image deployment

HVAC Electrical Technician, *Airflow Cooling & Heating Inc.*

June – Sept 2023

Troubleshoot electrical faults in HVAC systems; installed thermostat control systems and wired per electrical code.

Freelance Electronics Technician, *Self-Employed*

Jan 2020 – Present

Served 50+ clients; console mods, firmware flashing, custom PC builds, component-level SMD repair; 5.0/5.0 rating.

Leadership & Activities

Fordham Rocket Propulsion Lab, Co-Founder (Sept 2024 – May 2026): Propulsion design, thrust calculations, trajectory modeling, safety analysis; Fusion 360 fabrication | **Fordham Physics Society** (Sept 2023 – May 2026)

Physics & Math Tutor, Knack (June 2024 – Present): 5.0/5.0 rating across 50+ sessions